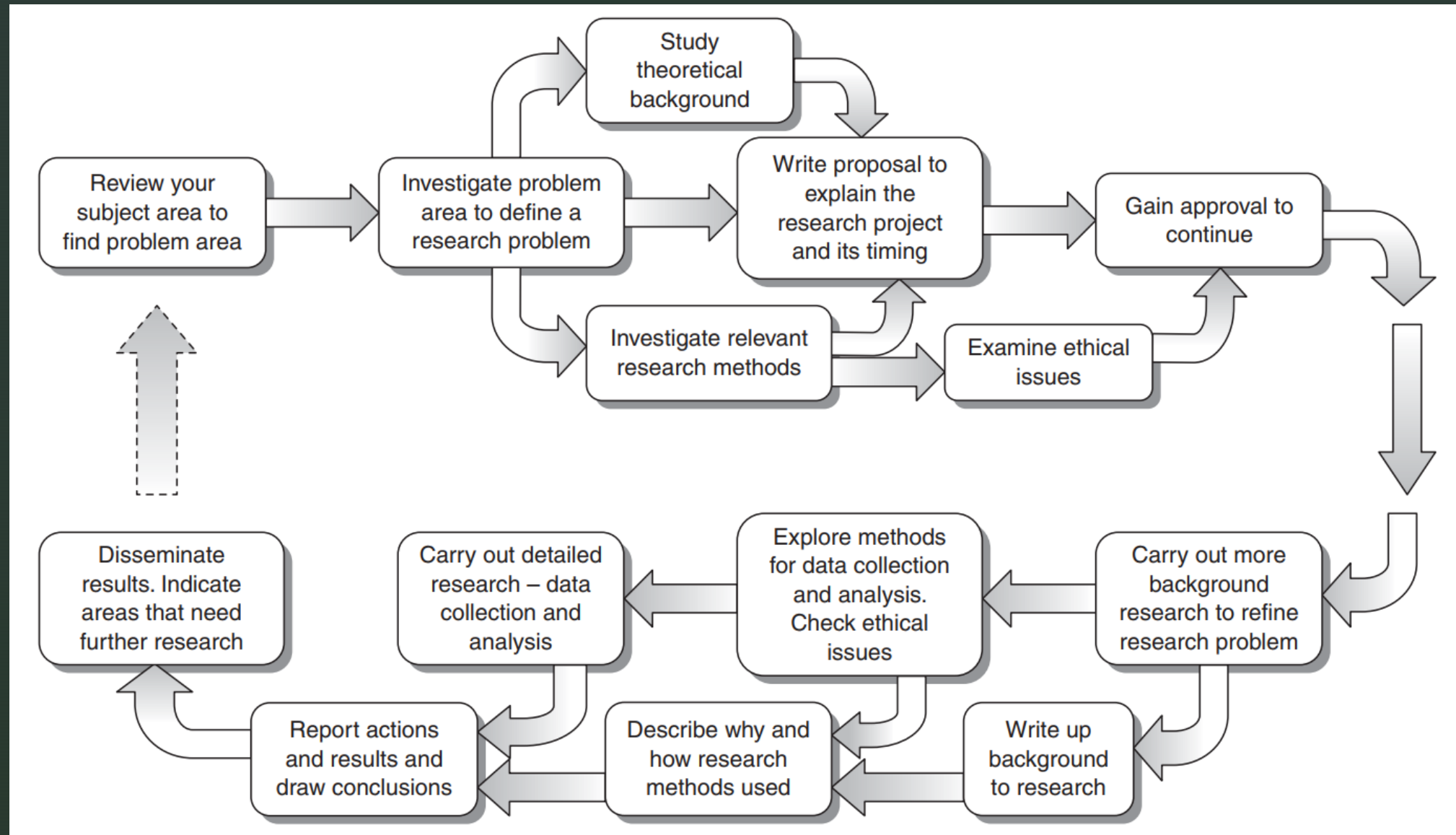


Unit 02: STRUCTURING YOUR RESEARCH PROJECT

Scientific Research

STRUCTURE of a RESEARCH PROJECT



STAGES of a RESEARCH PROJECT

- Formation of a precise **question (hypothesis)**, the answer to which will satisfy the aim of the research.
- Development of a detailed **understanding**, through reading and critical analysis of scientific literature and other resources. (**literature review**)
- **Gathering of evidence** that relates to the question, through experiment, analysis, or theory. These are intended to support or disprove the hypothesis underlying the question.
- Linking of the question and evidence with an **argument**, that is, a chain of reasoning. (**analysis of evidence**)
- Description of the work in a **publication**.

PURPOSE of RESEARCH

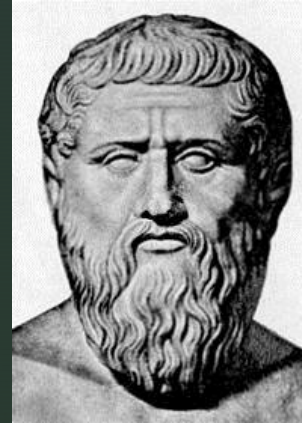
- A research project leads to papers and theses. These represent our store of accepted scientific knowledge.
- It is about having impact; by **creating new knowledge**, successful research changes the practices and understandings of other scientists.
- Our work must be adopted in some way by others if it is to be of value.
- Publication is not an end in itself but is **part of an ongoing collaborative enterprise**.

- A research usually begins with **curiosity**.
 - A student attending a lecture wonders why search engines do not provide better spelling correction.
 - A researcher investigating external sorting is at a seminar on file compression and ponders whether one could be of benefit to the other.
- Tea-room arguments are a rich source of seed ideas.
 - Someone claims that a researcher's idea is unworkable, and a listener starts to turn over the arguments. What makes it unworkable? How might those issues be addressed?

BEGINNINGS

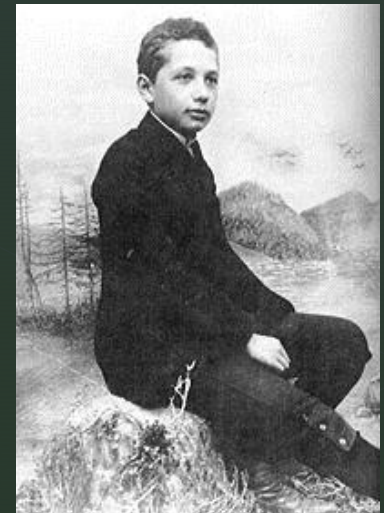
“*Philosophy begins in **wonder**.*”

Plato



“*To me it suffices to **wonder** at these secrets and to attempt humbly to grasp with my mind a mere image of the lofty structure of all that there is.*”

Albert Einstein



- The first step is **subjective**. We choose to explore ideas that:
 - seem likely to succeed
 - are intriguing
 - have the potential to lead to something new
- But the final outcome must be an **objective** scientific report.

CHOOSING RESEARCH TOPIC

- Choose a topic that is interesting to you.
- Sometimes, should focus on questions where work can be done, rather than where we would like to work.
- The fact that a topic is in a fashionable area should be at most a minor consideration.
- Don't tackle a problem that is too large to resolve.
- Select a topic that make use of your strengths.

STRENGTHS and TOPICS

STRENGTHS	TOPICS
<i>Statistical</i>	Identify properties of Web pages that are useful in determining whether they are good answers to queries.
<i>Mathematical</i>	Prove that the efficiency of index construction has reached a lower bound in terms of asymptotic cost.
<i>Analytical</i>	Quantify bottlenecks in query processing and relate them to properties of computers and networks.
<i>Algorithmic</i>	Develop and demonstrate the benefit of a new index structure.
<i>Representational</i>	Propose and evaluate a formal language for capturing properties of image, video, or audio to be used in search.
<i>Behavioral</i>	Quantify the effect on searchers of varying the interface.
<i>Social</i>	Link changes in search technology to changes in queries and user demographic.

RESEARCH PLANNING

- Having a series of **milestones** is critical to the success of a project.
- Milestones can always be identified, because (obviously, I hope!) any substantial project can be meaningfully described as a collection of **smaller activities**.
- Milestones do not necessarily have to be completed in turn.
- We should plan research with the following attitude: what evidence must we collect to convince a skeptical reader that the results are correct?
- The next step is to estimate the dates at which milestones should be reached. Keep in mind that everything takes longer than planned for.

RESEARCH PLANNING

- One approach of setting milestones is to explicitly consider what is needed at the end, then reason backwards. The final thing required is the write-up in the form of a thesis, paper, or report; so you need to plan in terms of the steps necessary to produce the write-up.
- As an example, consider research that is expected to have a substantial experimental component; the write-up is likely to involve a background review, explanations of previous and new algorithms, descriptions of experiments, and analysis of outcomes. Completion of each of these elements is a milestone.

HYPOTHESES

- The first stage of a research project involve choice of interesting topics and then formulating a hypothesis.
- A hypothesis is a statement of belief about how something works, or interacts, or behaves.
- Many hypotheses concern some aspect of the physical world: whether Web search can be used effectively by children or predict how well a service will respond to increasing load.
- Other hypotheses are constructive. For example, we propose new technologies and explore their limitations and feasibility or propose theorems that imply that there may be new solutions to long-standing algorithmic problems.
- The remaining part of this lecture concerns hypotheses and research questions, and how we use evidence to confirm or disprove them.

EXAMPLE of a RESEARCH PROJECT

- A researcher investigating algorithms might speculate as to whether it is possible to make better use of the cache on a CPU to reduce computational costs.
- Preliminary investigation might lead to the **hypothesis** that a tree-based structure with poor memory locality will be slower in practice than an array-based structure with high locality, despite the additional computational cost.
- The hypothesis suggests the **research question** of whether a particular sorting algorithm can be improved by replacing the tree structure with the array structure.
- The **phenomenon** that should be observed if the hypothesis is correct is a trend: for example, as the number of items to be sorted is increased, the tree-based method should increasingly show a high rate of cache misses compared to the array-based method.
- The **evidence** is the number of cache misses for several sets of items to be sorted. Alternatively, external evidence might be used, such as changes in execution time as the volume of data changes.

FEATURES of HYPOTHESES

- Be specific, **precise**, and unambiguous. A hypothesis should state what the limits on the conclusions will be so that it is **testable**.
- Example:
 - Suppose P-lists are a well-known data structure used for a range of applications, in particular as an in-memory search structure that is fast and compact. A new data structure called the Q-list is developed. The two structures to have the same asymptotic complexity in both space and time.
 - An incorrect hypothesis: *Q-lists are superior to P-lists.*
 - A more precise and testable hypothesis: *As an in-memory search structure for large data sets, Q-lists are faster and more compact than P-lists.* We assume there is a skew access pattern, that is, that the majority of accesses will be to a small proportion of the data.

FEATURES of HYPOTHESES

- A hypothesis must be capable of **falsification**. Examples of vague hypotheses:
 - *Q-list performance is comparable to P-list performance.*
 - *Our proposed query language is relatively easy to learn.*
- A hypothesis should lead to **predictive** results. In general the point of experimentation is to confirm models or theories, which can then be used to predict future behavior.

FEATURES of HYPOTHESES

- It may be necessary to **refine** a hypothesis after initial testing; indeed, much of scientific progress can be viewed as refinement and development of hypotheses to fit new observations.
- However, the hypothesis **should not follow the experiments**. There is a vast difference between an observation such as “the algorithm worked on our data” and a tested hypothesis such as “the algorithm was predicted to work on any data of this class, and this prediction has been confirmed on our data”.
- Where two hypotheses fit the observations equally well and one is clearly **simpler** than the other, the simpler should be chosen.

DEFENDING HYPOTHESESE

- One component of a strong research paper is the testing of the hypothesis and the presentation of the **supporting evidence**.
- In presenting the hypothesis, you need to construct an **argument** relating your hypothesis to the evidence.
- For example:
 - The hypothesis “the new range searching method is faster than previous methods” might be supported by the evidence “range search amongst n elements requires $2 \log_2 \log_2 n + c$ comparisons”.
 - The connecting argument may be “results for previous methods indicated an asymptotic cost of $\Theta(\log n)$ ”
- It is the role of the connecting argument to show that the evidence does indeed support the hypothesis, and to show that conclusions have been drawn correctly.

DEFENDING HYPOTHESES

- In constructing an argument, it can be helpful to imagine yourself defending your hypothesis to a colleague. That is, **raising objections and defending yourself against them** is a way of gathering the material that is needed to convince the reader that your argument is correct.
- For example, starting from the hypothesis that “the new string hashing algorithm is fast because it doesn’t use multiplication or division” you might debate as follows:
 - I don’t see why multiplication and division are a problem.
On most machines they use several cycles or may not be implemented in hardware at all. The new algorithm instead uses two exclusive-or operations per character and a modulo in the final step. I agree that for pipelined machines with floating-point accelerators the difference may not be great.
 - Modulo isn’t always in hardware either
True, but it is only required once.
 - So there is also an array lookup? That can be slow.
Not if the array is cache-resident.

DEFENDING HYPOTHESES

- Always **consider the possibility that your hypothesis is wrong**.
For example:
 - Are there any improbable consequences if the hypothesis is true?
 - For a hypothesis that displaces or contradicts some currently held belief, is the contradiction such that the belief can only have been held out of stupidity?
 - Does the hypothesis cover all of the observations explained by the current belief?
- A corollary is that the stronger your intuitive liking for a hypothesis, the more rigorously you should test it—that is, attempt to confirm it or disprove it—rather than twist results, and yourself, defending it.

- There are two basic stages to an argument: the **premises**, which are statements in the form of propositions or assertions which form the basis of the argument (this can be seen as the evidence) and the **conclusion**, which is a proposition that expresses the inference drawn by logical steps from the original premises.
- Arguments are based on logical reasoning:
 - Inductive reasoning
 - Deductive reasoning
 - The hypothetico-deductive method or scientific method is a further development of logical reasoning based on the principle that we can never be completely sure of any premises or conclusions that we make, but we can be more confident about some than others.
- It is important to be aware of the type of reasoning that you will use to make your argument, or if you are reading through the research literature, what type of reasoning is used by each of the writers.

TESTING ARGUMENTS

- As well as constructing arguments to support your research conclusions, it is also necessary to recognize and scrutinize other people's arguments, particularly those on whose work you are basing your research.
- There are techniques that can be used to detect the valid from the fallacious arguments. **Fallacies** fall into two main categories: formal and informal.

TESTING ARGUMENTS

- **Formal fallacies** are those where the logical structure underpinning the argument is faulty in some way. There are many ways that logical structures can be faulty.
- For one example, consider the following simple argument:
No lawyers are criminals.
John is not a lawyer.
Therefore, John is a criminal.
- This makes a hidden assumption that all people who are not lawyers are criminals. This is not one of the premises.

TESTING ARGUMENTS

- **Informal fallacies** also come in many guises. These are misleading not so much because the logical steps are incorrect, but they make false analogies, use emotional and misleading claims, and jump to unjustified conclusions on evidence that is skimpy, irrelevant or untrue.
- The following example makes false analogies:

The ship of government, like any ship, works best when there is a strong captain in charge of it. That is why government by dictatorship is more effective.
- Is a ship really sufficiently like a government to make this claim?

CONSTRUCTING ARGUMENTS

- There are only two basic aims of argumentation. One is to argue for a statement by providing evidence that will support it. The other is to refute a statement by providing evidence that undermines it.
- To argue in support of a concluding statement, you need to demonstrate that the premises are true
 - Evidence in the form of data confirmation or agreement amongst experts will help to substantiate the strength of your claim.
 - In research, evidence will often be new data generated by the project, so the reliability of sources and the methods of data collection must be clearly explained and justified.
 - You then must employ a sound logical progression of steps to bring you to the conclusions. The conclusions must be based exclusively on the premises.

CONSTRUCTING ARGUMENTS

- To refute a concluding statement requires you to show that the conclusion is false or untrue.
 - This can be done by challenging the logic of the argument, challenging the truth of evidence on which the argument is based, or questioning the relevance or completeness of the evidence.
 - Another way to challenge an argument is to produce counter-examples that are deployed against either the premises or/and conclusions. This will undermine the universality of the argument i.e. demonstrating that it does not hold in all cases.
 - Alternatively, you might be able to demonstrate that the conclusions drawn will lead to absurdity if taken far enough.

CONSTRUCTING ARGUMENTS



- Normally, the problem is divided down into smaller components requiring separate investigations.
- Conclusions about these components will occur in various sections of the analysis part of the report or dissertation.
- Skill is then required to gather these up at the end in the concluding section to fit them together into a 'mosaic' that will present the complete picture of the conclusion to the main problem.

FORMS of EVIDENCE

- There are, broadly speaking, four kinds of evidence that can be used to support a hypothesis: proof, modelling, simulation, and experiment.
- **Proof.** A proof is a formal argument that a hypothesis is correct (or wrong).
 - It is a mistake to suppose that the correctness of a proof is absolute—confidence in a proof may be high, but that does not guarantee that it is free from error; it is common for a researcher to feel certain that a theorem is correct but have doubts about the mechanics of the proof.

FORMS of EVIDENCE

- **Model.** A model is a mathematical description of the hypothesis (or some component of the hypothesis, such as an algorithm whose properties are being considered) and there will usually be a demonstration that the hypothesis and model do indeed correspond.
 - In choosing to use a model, consider how realistic it will be, or conversely how many simplifying assumptions need to be made for analysis to be feasible.

FORMS of EVIDENCE

- **Simulation.** A simulation is usually an implementation or partial implementation of a simplified form of the hypothesis, in which the difficulties of a full implementation are sidestepped by omission or approximation.
 - A simulation is a “white coats” test: artificial, isolated, and conducted in a tightly controlled environment.
 - A great advantage of a simulation is that it provides parameters that can be smoothly adjusted, allowing the researcher to observe behavior across a wide spectrum of inputs or characteristics.
 - However, with a simulation there is always the risk that it is unrealistic or simplistic, with properties that mean that the observed results would not occur in practice.

FORMS of EVIDENCE

- **Experiment.** An experiment is a full test of the hypothesis, based on an implementation of the proposal and on real—or highly realistic—data.
 - In an experiment there is a sense of *really doing it*, while in a simulation there is a sense of *only pretending*. For example, artificial data provides a mechanism for exploring behavior, but corresponding behavior needs to be observed on real data if the outcomes are to be persuasive.
 - Ideally an experiment should be conducted in the light of predictions made by a model, so that it confirms some expected behavior. An experiment should be severe; seek out tests that seem likely to fail if the hypothesis is false and explore extremes.

USE of EVIDENCE

- Different forms of evidence can be used to confirm one another, with say a simulation used to provide further evidence that a proof is correct.
- When choosing whether to use a proof, model, simulation, or experiment as evidence, consider how convincing each is likely to be to the reader. Select a form of evidence, not so as to keep your own effort to a minimum, but to **be as persuasive as possible**.
- Your written work is the one chance to persuade readers to accept the ideas, and they will only do so if the evidence and arguments are **complete and convincing**.

APPROACHES to MEASUREMENT

- The purpose of experimentation is to take **measurements** that can be **used as evidence**.
- Measurements can be quantitative, such as number or duration or volume—the speed of a system, say, or an algorithm's efficiency relative to a baseline.
- Measurements can also be qualitative, such as an occurrence or difference—whether an outcome was achieved, or whether particular features were observed.

APPROACHES to MEASUREMENT

- As you develop your research questions, then, you should ask *what is to be measured?* and *what measures will be used?*
- For example, when examining an algorithm, will it be measured by execution time? And if so, what mechanism will be used to measure it?
- This question can be tricky to answer for a single-threaded process running on a single machine. For a distributed process using diverse resources across a network, there probably is no perfect answer, only a range of choices with a variety of flaws and shortcomings, each of which needs to be understood by you and by your readers.

TYPICAL COMPUTER SCIENCE SCENARIO

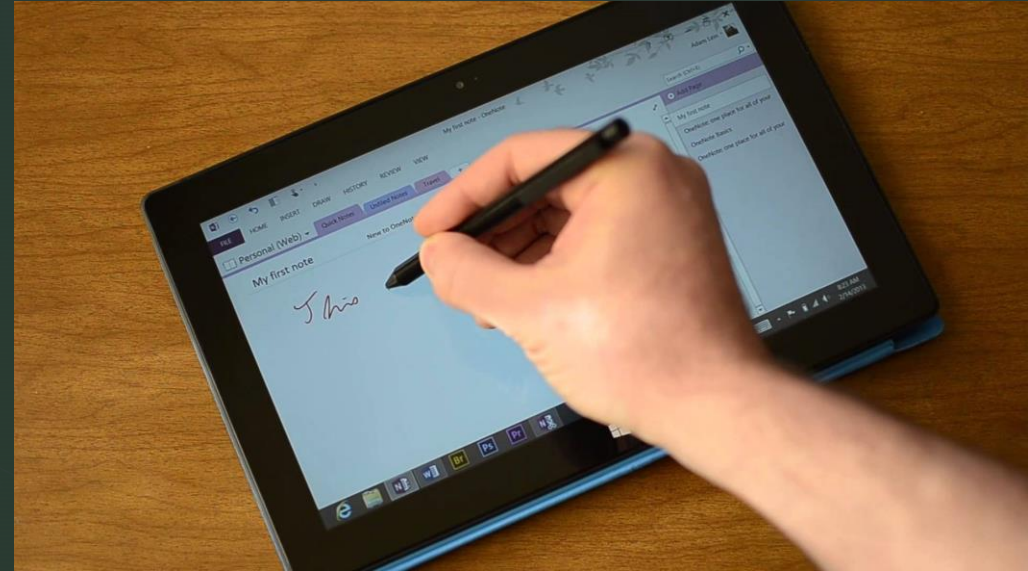
- A particular task needs to be solved by a software system
- This task is currently solved by an existing system (a baseline)
- You propose a new, in your opinion, better system.
- You argue why your proposed system is better than the baseline.
- You support your arguments by providing evidence that your system indeed beats the baseline.

EXAMPLE of a CS RESEARCH PROJECT



Text entry on a Tablet PC

- A. Handwriting recognition
- B. Software keyboard (baseline)



EXAMPLE of a CS RESEARCH PROJECT



- Why this experiment?
 - Despite decades of research there is no empirical data of text entry performance of handwriting recognition.
 - An inappropriate study of handwriting (sans recognition) from 1967 keeps getting cited in the literature, often through secondary or tertiary sources (handbooks, etc.)
 - Based on these numerous citations in research papers, handwriting recognition is perceived to be rather slow
 - However, there is no empirical evidence that supports this claim

EXAMPLE of a CS RESEARCH PROJECT

- Controlled experiments and hypotheses
 - A controlled experiment tests the validity of one or more hypothesis
 - Here we will consider the simplest case:
 - One method vs. another method
 - Each method is referred to as a *condition*
 - The null hypothesis H_0 states there is no difference between the conditions
 - Our hypothesis H_1 states there *is* a difference between the conditions
 - To show a statistically significant difference the null hypothesis H_0 needs to be rejected

EXAMPLE of a CS RESEARCH PROJECT



Why this baseline?

- The software keyboard is well understood
 - Many empirical studies of their performance
 - Also exists expert computational performance models
- The software keyboard is the *de-facto* standard text entry method on tablets.
- The literature compares handwriting recognition text entry performance against measures of the software keyboard.

EXAMPLE of a CS RESEARCH PROJECT

- We let participants write phrases using either:
 - Software keyboard (baseline)
 - Handwriting recognition
 - That is, we manipulate the *input method*
- We measure:
 - Entry rate in words-per-minute
 - Error rate in number of written characters that do not match the stimulus

EXAMPLE of a CS RESEARCH PROJECT



Explaining what you did

- An experiment needs to be reproducible by others.
- It is your responsibility to ensure that you explained your experimental procedure in enough detail.
- Choices made in the experimental design needs to be motivated.
- This part of a research paper is typically referred to as the *Method* section.
 - Participants
 - Apparatus
 - Procedure

EXAMPLE of a CS RESEARCH PROJECT

Participants

- We recruited **12 volunteers** from the **university campus**. We intentionally wanted a rather broad sample and recruited participants from many different departments with many different backgrounds. **Six were men and six were women**. Their **ages ranged between 22-37** (mean = 27, sd = 4). Participants were **screened** for dyslexia and repetitive strain injury (RSI). **Seven** participants were **native English speakers** and **five** participants had **English as their second language**. No participant had used a handwriting recognition interface before. One participant had used a software keyboard before. No participant had regularly used a software keyboard before. Participants **were compensated £10 per session**.

EXAMPLE of a CS RESEARCH PROJECT

Apparatus

- We used a Dell Latitude XT Tablet PC running Windows Vista Service Pack 1. The 12.1" color touch-screen had a resolution of **1280 × 800 pixels** and a physical screen size of **261 × 163 mm**. Participants used a **capacitance-based** pen to write directly onto the screen in both conditions.
- Both the handwriting recognizer and the software keyboard were docked to the **lower part of the screen**. The dimensions of the software keyboard were 1266 × 244 pixels and 257 × 50 mm. The dimensions of the handwriting recognizer writing area measured 1266 × 264 pixels and 257 × 55 mm.

EXAMPLE of a CS RESEARCH PROJECT

Apparatus

- The handwriting recognizer was **configured to learn and adapt** to participants' handwriting style (the **default setting** on Windows Vista). Each participant performed the experiment in a **separate user account** on the machine to ensure handwriting adaptation was carried out on an individual basis. There was a **potential confound** in enabling handwriting adaptation since it caused the system, as well as the user, to learn as a function of usage. In the interest of **external validity** we enabled adaptation **since in actual use** users would most likely have adaptation turned on.

EXAMPLE of a CS RESEARCH PROJECT

Apparatus (motivating your choice)

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EXAMPLE of a CS RESEARCH PROJECT

Procedure

- The experiment consisted of **one introductory** session and **ten testing** sessions. In the introductory session the experimental procedure was explained to the participants. Participants were shown how to use the software keyboard and the handwriting recognizer, including demonstrations of how to correct errors.

...

Each **testing session lasted slightly less than one hour**. Testing sessions were **spaced at least 4 hours** from each other, and subsequent testing sessions were **maximally separated by two days**. In each testing session participants **did both conditions** (software keyboard and handwriting recognition). The **order of the conditions alternated between sessions** and the **starting condition was balanced across participants**. **Each condition lasted 25 minutes**. Between conditions there was a brief break. Participants were also instructed that they could rest at any time after completing an individual phrase.

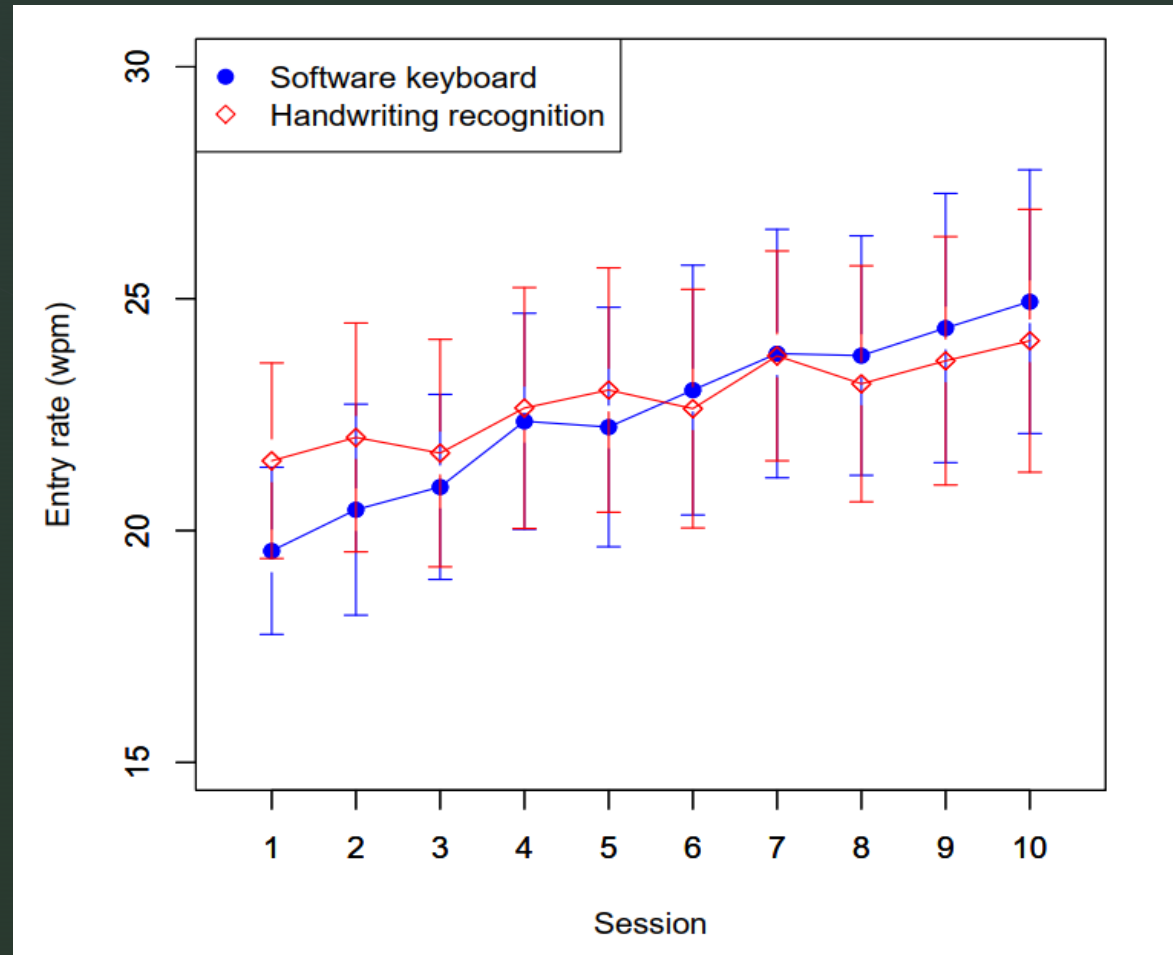
EXAMPLE of a CS RESEARCH PROJECT

Procedure

- In each condition participants were **shown a phrase** drawn from the **phrase set** provided by MacKenzie and Soukoreff [8]. Each participant had their **own randomized copy of the phrase set**. Participants were instructed to **quickly and accurately** write the presented phrase using either the software keyboard or the handwriting recognizer. Participants were **instructed to correct any mistakes they spotted in their text**. In the handwriting condition we instructed participants to **write using their preferred style of handwriting** (e.g. printed, cursive or a mixture of both). After they had written the phrase, they pressed a Submit button and the next phrase was displayed. The Submit button was a rectangular button measuring 248×16 mm. It was placed 9 mm above the keyboard and handwriting recognizer writing area.

EXAMPLE of a CS RESEARCH PROJECT

Results



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